

EE4615 Digital IC II

Instruction Manual

Last revised April 2026

This document is a guide for the setting up and accessing Cadence for the course EE4615 Digital IC Design II. The technology used in this course is updated to TSMC 180 nm due to license issues in ADE.

Cadence Setup:

We access the Cadence design tool from a remote server, namely ee4615.ewi.tudelft.nl. Most students will connect over SSH using MobaXterm. The procedure for setting up is listed below:

1. Open MobaXterm and type `ssh -X <netid>ee4615.ewi.tudelft.nl`. (The username is your netid)
2. Enter your password in the terminal
3. Create a directory 'tsmcBCD' using the command `mkdir tsmcBCD`
4. Change home directory using `cd tsmcBCD`
5. Type `/opt/ei/DK/tsmc/oa180/mini018BCDg2/216A/et4382/start.ee4615` in the terminal
6. Type `source sourceme.ee4615` in the terminal
7. Type `virtuoso &` to start up cadence

These steps need to be done only once. From next time onwards, to start up cadence follow the below steps.

1. Open MobaXterm and type `ssh -X <netid>ee4615.ewi.tudelft.nl`. (The username is your netid)
2. Enter your password in the terminal (If you already saved your password, skip this step.)
3. Change home directory using `cd tsmcBCD`
4. Type `source sourceme.ee4615` in the terminal
5. Type `virtuoso &` to start up cadence

NOTE: While creating your library, make sure to link your library to tsmc18.

While opening schematics and analog environment (ADE Explorer) you may face warnings regarding license, please click OK or Always.

For design of oscillators and TDC, use transistor models `nmos2v`, `nmos5v`, `pmos2v` and `pmos5v`. Suggestion from us is that in case Cadence is slow, you can implement the initial design or the entire design in LTspice (Technology used is similar to the one used in Analog CMOS I and II so you can reuse that setup files).